

Lecture 10: The Matrix (continued)

Instructors:

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[Based on slides for „Decoding the Matrix“]



WEEK

FIVE



Agenda

Chapter 4

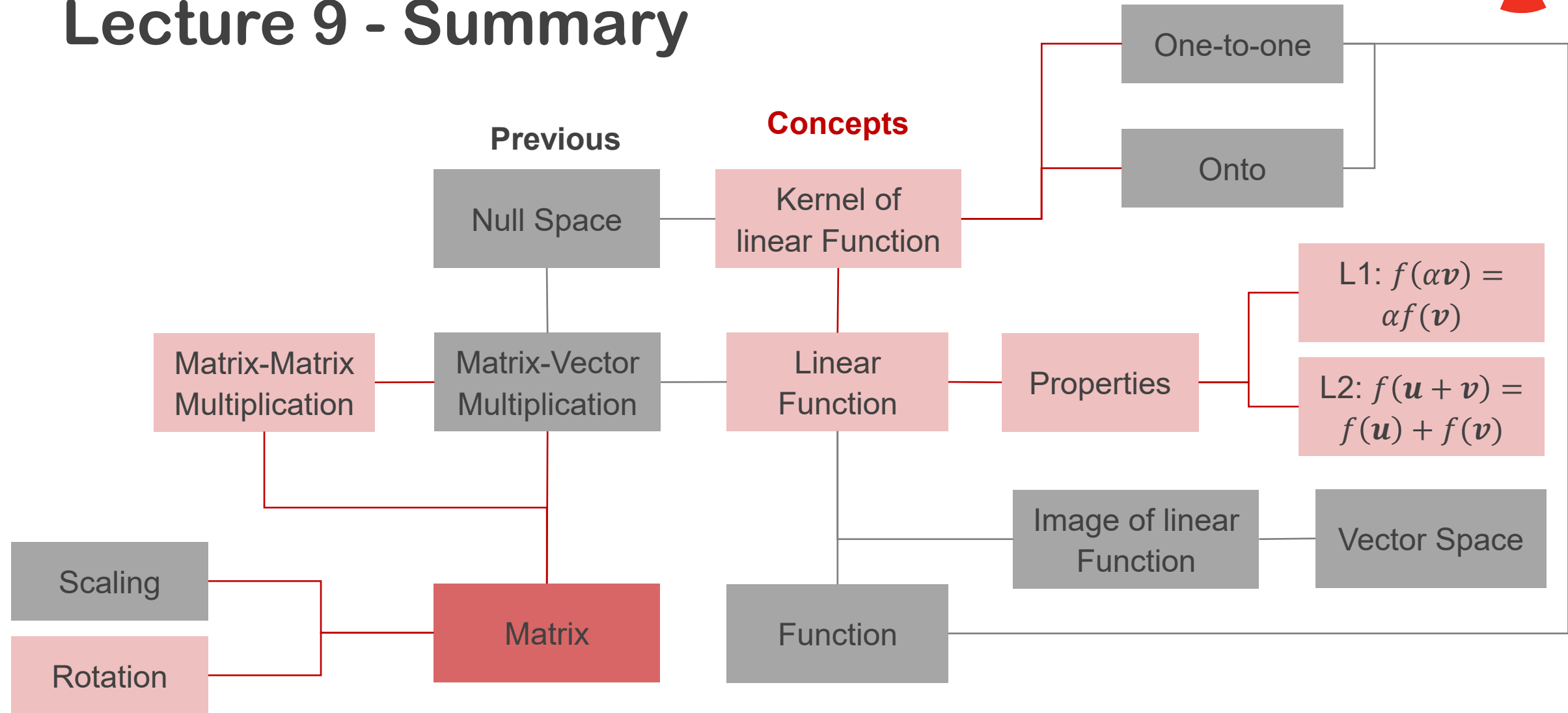
Lecture 10:

- Column Vector
- Row Vector
- Inner Product
- Outer Product
- Function Composition
- Matrix Determinant – Meaning
- Matrix Determinant – 3-by-3
- Matrix Inverse
- Product of Invertible Matrices





Lecture 9 - Summary



Vector-Vector Multiplication

(General) Linear System

$$\begin{array}{rcl} \mathbf{a}_1 \cdot \mathbf{x} & = & \beta_1 \\ & \vdots & \\ \mathbf{a}_m \cdot \mathbf{x} & = & \beta_m \end{array}$$

Affine Space (or empty)

Homogeneous Linear System

$$\begin{array}{rcl} \mathbf{a}_1 \cdot \mathbf{x} & = & 0 \\ & \vdots & \\ \mathbf{a}_m \cdot \mathbf{x} & = & 0 \end{array}$$

Vector Space

If a linear system has a solution $\mathbf{u}_1$ then that solution is unique if the only solution to the corresponding homogeneous linear system is $\mathbf{0}$.

Matrix-Vector Multiplication

Matrix-Vector Equation

$$\underbrace{\begin{bmatrix} \mathbf{a}_1 \\ \mathbf{a}_2 \\ \vdots \\ \mathbf{a}_m \end{bmatrix}}_A * \mathbf{x} = \begin{bmatrix} \beta_1 \\ \beta_2 \\ \dots \\ \beta_m \end{bmatrix}$$

(Matrix-Vector Equation) for Null Space

Null space of a matrix A is $\{\mathbf{u} : A * \mathbf{u} = \mathbf{0}\}$.

$$\begin{bmatrix} \mathbf{a}_1 \\ \mathbf{a}_2 \\ \vdots \\ \mathbf{a}_m \end{bmatrix} * \mathbf{u} = \begin{bmatrix} 0 \\ 0 \\ \dots \\ 0 \end{bmatrix}$$

Vector Space

If $\mathbf{u}_1$ is a solution to $A * \mathbf{x} = \mathbf{b}$ then solution set is $\mathbf{u}_1 + \mathcal{V}$ where $\mathcal{V} = \text{Null } A$

Linear Functions

(implemented via Matrix-Vector Mult.)



Linear Function

(with L1 and L2 Properties)

$$f(\mathbf{x}) = A * \mathbf{x}.$$

(Linear function) for Kernel

Kernel of a linear function f is

$$\{\mathbf{v} : f(\mathbf{v}) = \mathbf{0}\}$$

$$f(\mathbf{v}) = M * \mathbf{v} = \begin{bmatrix} 0 \\ \dots \\ 0 \end{bmatrix}$$

For a function $f(\mathbf{x}) = M * \mathbf{x}$,

Vector Space

$$\text{Ker } f = \text{Null } M$$

If $\text{Ker } f$ is trivial (i.e. if $\text{Null } A$ is trivial) then a vector $\mathbf{b}$ is the image under f of at most one vector.

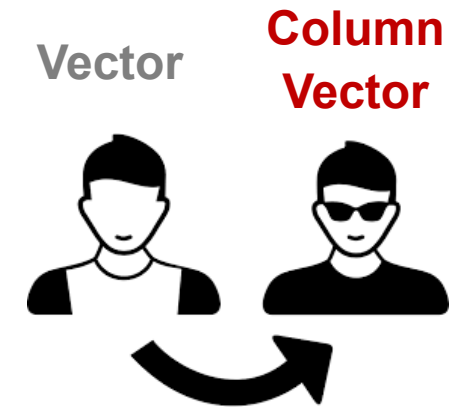


Column & Row Vector (I)

In contrast to the book, we have directly considered the **orientation** of vectors when performing calculations

→ Need to match e.g. $(2 \times 3) \times (3 \times 1) \rightarrow$ leads to (2×1) matrix

Convention: Interpret a vector **b** as a one-column matrix (“column vector”)

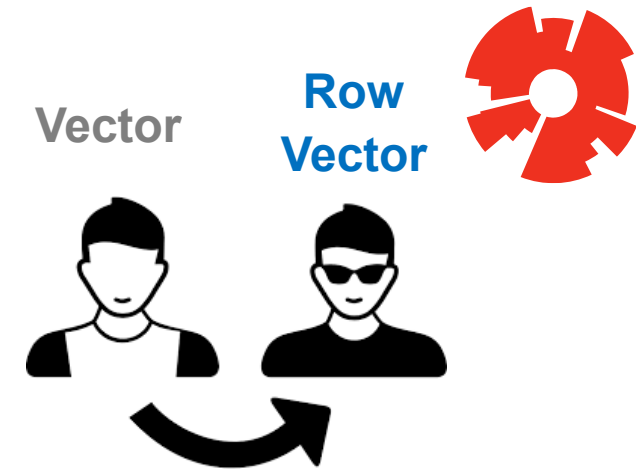


We often wrote this as $\begin{bmatrix} A \end{bmatrix} [1, 2, 3]^T$

To focus on the linear combination interpretation

Keep in mind that $\begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}$ and $[1, 2, 3]^T$ are the same

Column & Row Vector (II)



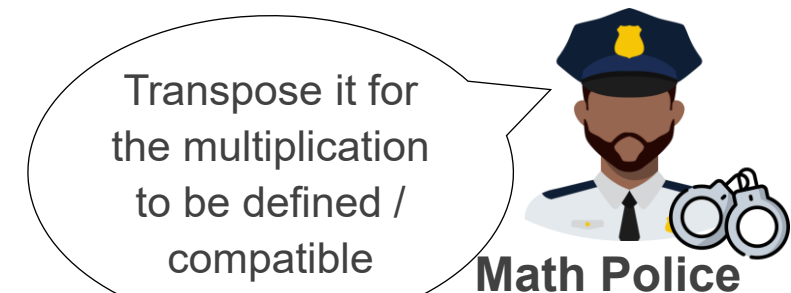
Convention: Similar to column vectors, we could interpret a vector as a **one-row-matrix** (“row vector”)

Since we assume that vectors are by **default column vectors** (unless otherwise specified)
 → having a row vector usually includes indicating that it was **transposed**

Use **transpose** to turn a column vector into a row vector: Suppose $b = \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}$

$$\mathbf{b}^T A = [1, 2, 3] * A = \begin{bmatrix} 1 & 2 & 3 \end{bmatrix} \begin{bmatrix} A \end{bmatrix}$$

(1×3) (3×2)





Matrix-Matrix Mult. - Inner Product

Let $\mathbf{u}$ and $\mathbf{v}$ be two D -vectors interpreted as matrices (column vectors).

Matrix-matrix product $\mathbf{u}^T \mathbf{v}$.

Example: $\begin{bmatrix} 1 & 2 & 3 \end{bmatrix} \begin{bmatrix} 3 \\ 2 \\ 1 \end{bmatrix} = \begin{bmatrix} 10 \end{bmatrix}$

- ▶ First “matrix” has one row.
- ▶ Second “matrix” has one column.
- ▶ Therefore product “matrix” has one entry.

So, we interpret vectors as matrices,
to be able to use matrix operations



$$1 \times n \quad n \times 1 \quad \rightarrow \quad 1 \times 1$$

By dot-product definition of matrix-matrix multiplication, that one entry is the dot-product of $\mathbf{u}$ and $\mathbf{v}$.

Sometimes called *inner product* of matrices.

However, that term has taken on another meaning, which we study later.



Matrix-Matrix Mult. - Outer Product

Another way to multiply vectors as matrices.

For any $\mathbf{u}$ and $\mathbf{v}$, consider $\mathbf{uv}^T$.

Example:

$$\begin{bmatrix} u_1 \\ u_2 \\ u_3 \end{bmatrix} \begin{bmatrix} v_1 & v_2 & v_3 & v_4 \end{bmatrix} = \begin{bmatrix} u_1 v_1 & u_1 v_2 & u_1 v_3 & u_1 v_4 \\ u_2 v_1 & u_2 v_2 & u_2 v_3 & u_2 v_4 \\ u_3 v_1 & u_3 v_2 & u_3 v_3 & u_3 v_4 \end{bmatrix}$$

For each element s of the domain of $\mathbf{u}$ and each element t of the domain of $\mathbf{v}$, the s, t element of $\mathbf{uv}^T$ is $\mathbf{u}[s] \mathbf{v}[t]$.

$$\leftarrow \begin{matrix} n \times 1 & 1 \times m & \rightarrow n \times m \end{matrix}$$

Called outer product of $\mathbf{u}$ and $\mathbf{v}$.



Matrix-Vector Definition of Matrix-Matrix Mult. (I)

Matrix-vector multiplication $\rightarrow$ **linear combination** of the columns of M

$$M * v = \begin{bmatrix} 1 & 2 \\ 0 & 1 \end{bmatrix} * [1, 2]^T = 1 * \begin{bmatrix} 1 \\ 0 \end{bmatrix} + 2 * \begin{bmatrix} 2 \\ 1 \end{bmatrix} = \begin{bmatrix} 5 \\ 2 \end{bmatrix}$$

Let's just write the **same equation** but displaying v as the column vector it is

$$M * v = \begin{bmatrix} 1 & 2 \\ 0 & 1 \end{bmatrix} * \begin{bmatrix} 1 \\ 2 \end{bmatrix} = 1 * \begin{bmatrix} 1 \\ 0 \end{bmatrix} + 2 * \begin{bmatrix} 2 \\ 1 \end{bmatrix} = \begin{bmatrix} 5 \\ 2 \end{bmatrix}$$

... so $M * v$ is the linear combination of the **columns of M** based on the elements in the **column vector v**



Matrix-Vector Definition of Matrix-Matrix Mult. (II)

$$M * v = \begin{bmatrix} 1 & 2 \\ 0 & 1 \end{bmatrix} * \begin{bmatrix} 1 \\ 2 \end{bmatrix} = 1 * \begin{bmatrix} 1 \\ 0 \end{bmatrix} + 2 * \begin{bmatrix} 2 \\ 1 \end{bmatrix} = \begin{bmatrix} 5 \\ 2 \end{bmatrix}$$

... so $M * v$ is the linear combination of the **columns of M** based on the elements in the **column vector v**

Matrix-matrix multiplication → Each column in resulting MA **linear combination** of the **columns of M** using the **columns of A**

$$M * A = \begin{bmatrix} 1 & 2 \\ 0 & 1 \end{bmatrix} * \begin{bmatrix} 1 & 3 \\ 2 & 4 \end{bmatrix} = \left[1 * \begin{bmatrix} 1 \\ 0 \end{bmatrix} + 2 * \begin{bmatrix} 2 \\ 1 \end{bmatrix}, 3 * \begin{bmatrix} 1 \\ 0 \end{bmatrix} + 4 * \begin{bmatrix} 2 \\ 1 \end{bmatrix} \right] = \begin{bmatrix} 5 & 11 \\ 2 & 4 \end{bmatrix}$$

Linear combination of **columns of M**
using **first column of A**

Linear combination of **columns of M**
using **second column of A**



Matrix-Vector Definition of Matrix-Matrix Mult. (III)

(you can double-check also with the dot-product that this is indeed the result)

$$M * A = \begin{bmatrix} 1 & 2 \\ 0 & 1 \end{bmatrix} * \begin{bmatrix} 1 & 3 \\ 2 & 4 \end{bmatrix} = \left[1 * \begin{bmatrix} 1 \\ 0 \end{bmatrix} + 2 * \begin{bmatrix} 2 \\ 1 \end{bmatrix}, 3 * \begin{bmatrix} 1 \\ 0 \end{bmatrix} + 4 * \begin{bmatrix} 2 \\ 1 \end{bmatrix} \right] = \begin{bmatrix} 5 & 11 \\ 2 & 4 \end{bmatrix}$$

Linear combination of **columns of M**
using **first column of A**

Linear combination of **columns of M**
using **second column of A**

By the matrix-vector definition of matrix-matrix multiplication, column j of MA is $M * (\text{column } j \text{ of } A)$.



Matrix-Matrix Mult. - Function Composition (I)

Remember
(Lecture 1)

Definition: For functions $f : A \longrightarrow B$ and $g : B \longrightarrow C$, the *functional composition* of f and g is the function $(g \circ f) : A \longrightarrow C$ defined by $(g \circ f)(x) = g(f(x))$

Corresponding to an $R \times C$ matrix A over a field $\mathbb{F}$, there is a function

$$f : \mathbb{F}^C \longrightarrow \mathbb{F}^R$$

namely the function defined by $f(\mathbf{y}) = A * \mathbf{y}$

Matrices A and $B \Rightarrow$ functions $f(\mathbf{y}) = A * \mathbf{y}$ and $g(\mathbf{x}) = B * \mathbf{x}$ and $h(\mathbf{x}) = (AB) * \mathbf{x}$

Matrix-Multiplication Lemma $f \circ g = h$

Welcome back old friend,
function composition



**The functions we focus on here are linear functions
($\rightarrow$ we can represent as a matrix-vector equation)**



Matrix-Matrix Mult. - Function Composition (II)

Example:

$$A = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix} \quad B = \begin{bmatrix} 1 & 0 \\ 1 & 1 \end{bmatrix} \quad AB = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 1 & 1 \end{bmatrix} = \begin{bmatrix} 2 & 1 \\ 1 & 1 \end{bmatrix}$$

$$f(\mathbf{y}) = A * \mathbf{y} \quad g(\mathbf{x}) = B * \mathbf{x}$$

$$f \circ g = f \left(\begin{bmatrix} 1 & 0 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} \right) = f \left(\begin{bmatrix} x_1 \\ x_1 + x_2 \end{bmatrix} \right) = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_1 + x_2 \end{bmatrix} = \begin{bmatrix} 2x_1 + x_2 \\ x_1 + x_2 \end{bmatrix}$$

$$h(\mathbf{x}) = (AB) * \mathbf{x} = \begin{bmatrix} 2 & 1 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 2x_1 + x_2 \\ x_1 + x_2 \end{bmatrix} \quad \text{so } f \circ g = h$$

→ composing two linear functions corresponds to their **matrix product**



Matrix-Matrix Mult. - Function Composition (III)



Proof: Let columns of B be $\mathbf{b}_1, \dots, \mathbf{b}_n$. By the matrix-vector definition of matrix-matrix multiplication, column j of AB is $A * (\text{column } j \text{ of } B)$.

For any n -vector $\mathbf{x} = [x_1, \dots, x_n]$,

$$g(\mathbf{x}) = B * \mathbf{x}$$

by definition of g

$$= x_1 \mathbf{b}_1 + \dots + x_n \mathbf{b}_n$$

by linear combinations definition

Therefore

$$f(g(\mathbf{x})) = f(x_1 \mathbf{b}_1 + \dots + x_n \mathbf{b}_n)$$

$$= x_1(f(\mathbf{b}_1)) + \dots + x_n(f(\mathbf{b}_n))$$

by linearity of f

$$= x_1(A * \mathbf{b}_1) + \dots + x_n(A * \mathbf{b}_n)$$

by definition of f

$$= x_1(\text{column 1 of } AB) + \dots + x_n(\text{column } n \text{ of } AB)$$

by matrix-vector def.

$$= (AB) * \mathbf{x}$$

by linear-combinations def.

$$= h(\mathbf{x})$$

by definition of h

QED

Same Proof but with some
(hopefully) helpful highlights



Matrix-Matrix Mult. - Function Composition (IV)



Proof: Let columns of B be $\mathbf{b}_1, \dots, \mathbf{b}_n$. By the matrix-vector definition of matrix-matrix multiplication, column j of AB is $A * (\text{column } j \text{ of } B)$. ← matrix-vector def. of Mat-Mat multiplication (few slides ago)
For any n -vector $\mathbf{x} = [x_1, \dots, x_n]$,

$$g(\mathbf{x}) = B * \mathbf{x}$$

by definition of g

$$= x_1 \mathbf{b}_1 + \dots + x_n \mathbf{b}_n$$

by linear combinations definition

Therefore

$$f(g(\mathbf{x})) = f(x_1 \mathbf{b}_1 + \dots + x_n \mathbf{b}_n)$$

$$= x_1 (f(\mathbf{b}_1)) + \dots + x_n (f(\mathbf{b}_n))$$

by linearity of f (L1 and L2)

$$= x_1 (A * \mathbf{b}_1) + \dots + x_n (A * \mathbf{b}_n)$$

by definition of f

$$= x_1 (\text{column 1 of } AB) + \dots + x_n (\text{column } n \text{ of } AB)$$

by matrix-vector def. ←

$$= (AB) * \mathbf{x}$$

by linear-combinations def.

$$= h(\mathbf{x})$$

by definition of h

QED



Matrix-Matrix Mult. - Associativity

Matrix-matrix multiplication corresponds to function composition.

Corollary: Matrix-matrix multiplication is associative:

$$(AB)C = A(BC)$$

Proof: Function composition is associative. QED

Example:

$$\begin{bmatrix} 1 & 0 \\ 1 & 1 \end{bmatrix} \left(\begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} -1 & 3 \\ 1 & 2 \end{bmatrix} \right) = \begin{bmatrix} 1 & 0 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} 0 & 5 \\ 1 & 2 \end{bmatrix} = \begin{bmatrix} 0 & 5 \\ 1 & 7 \end{bmatrix}$$

$$\left(\begin{bmatrix} 1 & 0 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix} \right) \begin{bmatrix} -1 & 3 \\ 1 & 2 \end{bmatrix} = \begin{bmatrix} 1 & 1 \\ 1 & 2 \end{bmatrix} \begin{bmatrix} -1 & 3 \\ 1 & 2 \end{bmatrix} = \begin{bmatrix} 0 & 5 \\ 1 & 7 \end{bmatrix}$$



Matrix Determinant – Calculation for 2 x 2

In 2D (for a **2x2 matrix** – only applies to square matrices)

$$A = \begin{bmatrix} a_{1,1} & a_{1,2} \\ a_{2,1} & a_{2,2} \end{bmatrix} \quad \longrightarrow \quad \det A = |A| = a_{1,1}a_{2,2} - a_{1,2}a_{2,1}$$

$$A = \begin{bmatrix} 2 & 0 \\ 0 & 2 \end{bmatrix}$$



$$\det A = 2 * 2 - 0 * 0 = 4$$

Positive

$$A = \begin{bmatrix} 0.5 & 2 \\ 2 & 0.5 \end{bmatrix}$$



$$\det A = 0.5 * 0.5 - 2 * 2 = -3.75$$

Negative

$$A = \begin{bmatrix} 2 & 1 \\ 2 & 1 \end{bmatrix}$$



$$\det A = 2 * 1 - 1 * 2 = 0$$

Zero

Non-zero

What does it tell
me ... and
WHY?

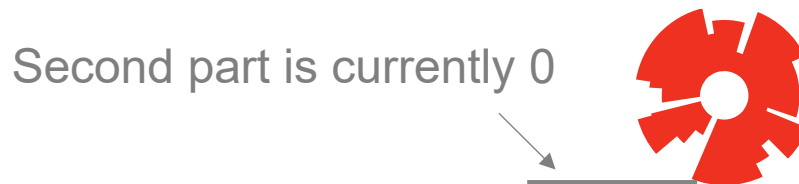
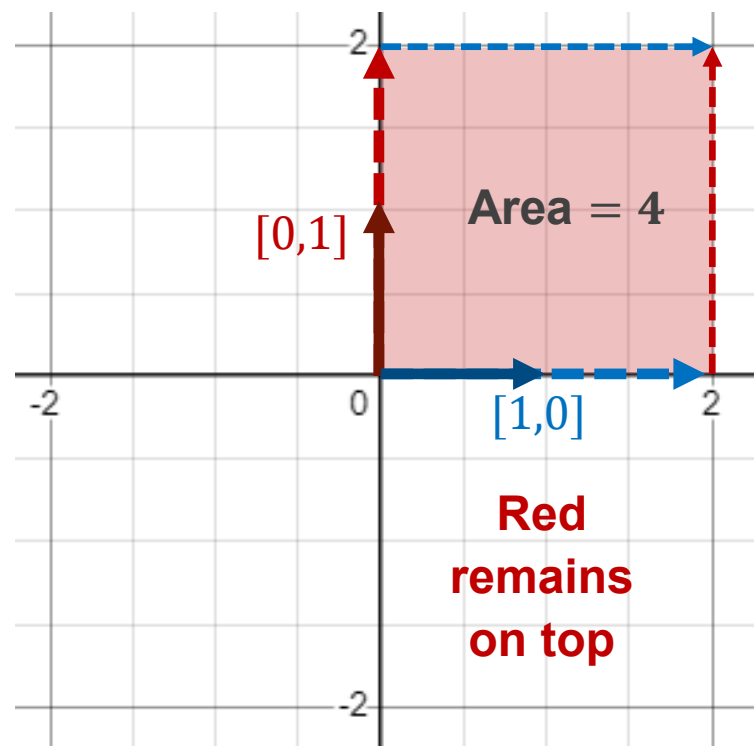
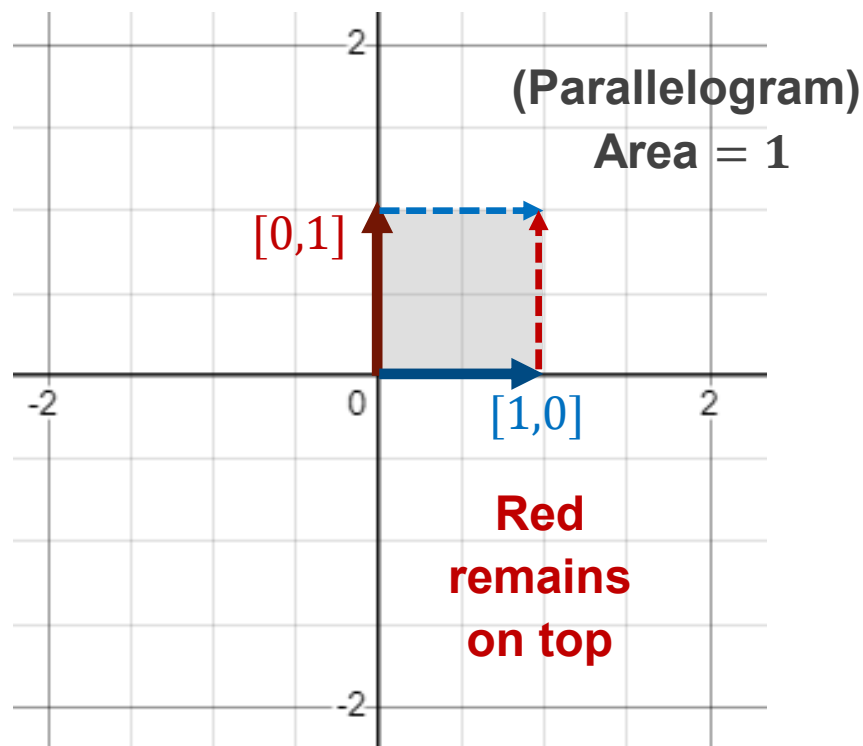


For Scaling

Matrix Determinant - Meaning (I)

Last time we already shaded the **area** but did not discuss it much – now we will!

Standard generators (no transformation yet)



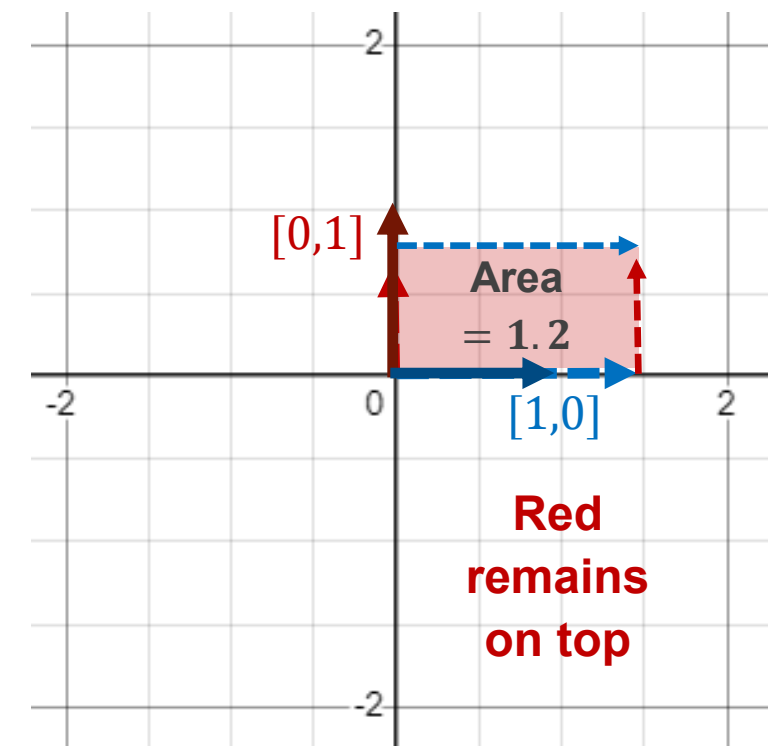
$$\det A = |A| = a_{1,1}a_{2,2} - a_{1,2}a_{2,1}$$

$$A = \begin{bmatrix} 2 & 0 \\ 0 & 2 \end{bmatrix}$$

$$\det A = 4 - 0 = 4$$

$$A = \begin{bmatrix} 1.5 & 0 \\ 0 & 0.8 \end{bmatrix}$$

$$\det A = 1.2 - 0 = 1.2$$





For shears

Matrix Determinant - Meaning (II)

$$A = \begin{bmatrix} 2 & 0.5 \\ 0.5 & 2 \end{bmatrix}$$

$$\det A = 4 - 0.25 = 3.75$$

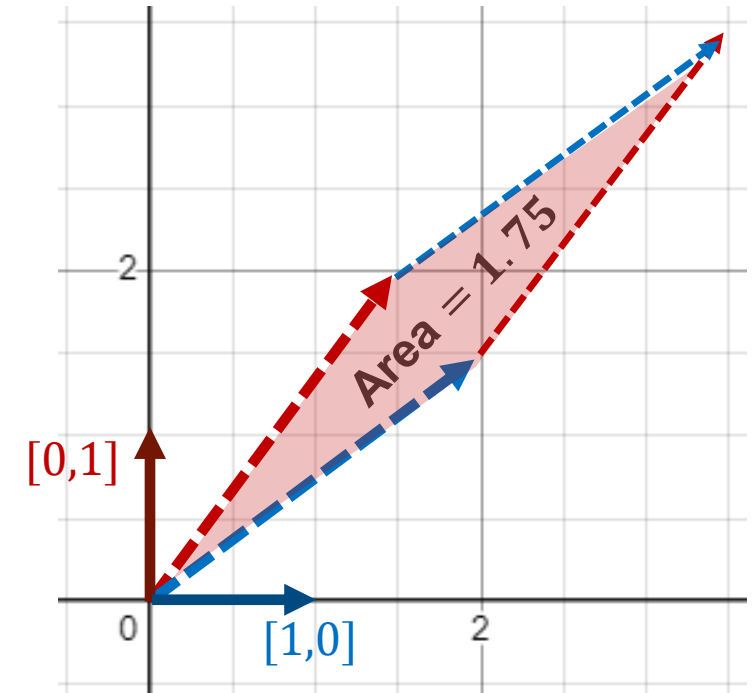
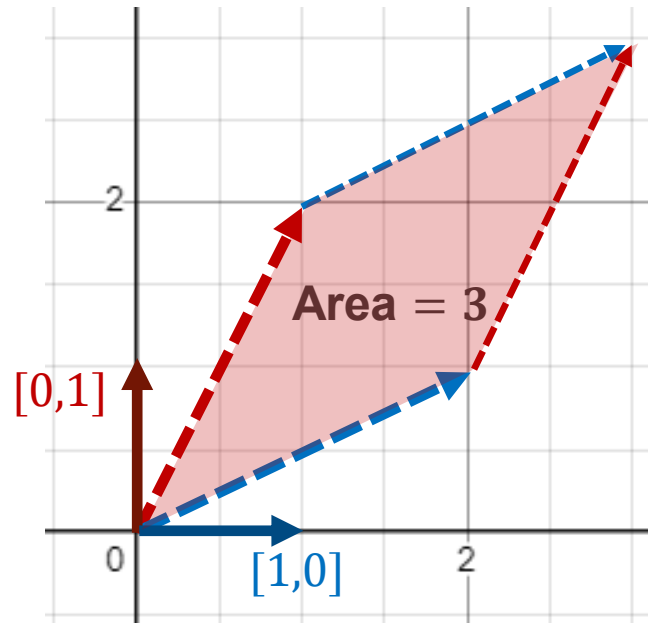
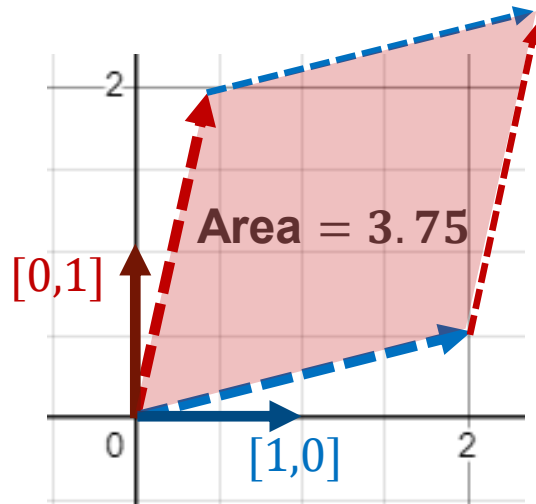
$$A = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}$$

$$\det A = 4 - 1 = 3$$

$$A = \begin{bmatrix} 2 & 1.5 \\ 1.5 & 2 \end{bmatrix}$$

$$\det A = 4 - 2.25 = 1.75$$

“shearing” / “a shear” → slanting / tilting of the parallelogram shape (because the off-diagonal entries are nonzero)





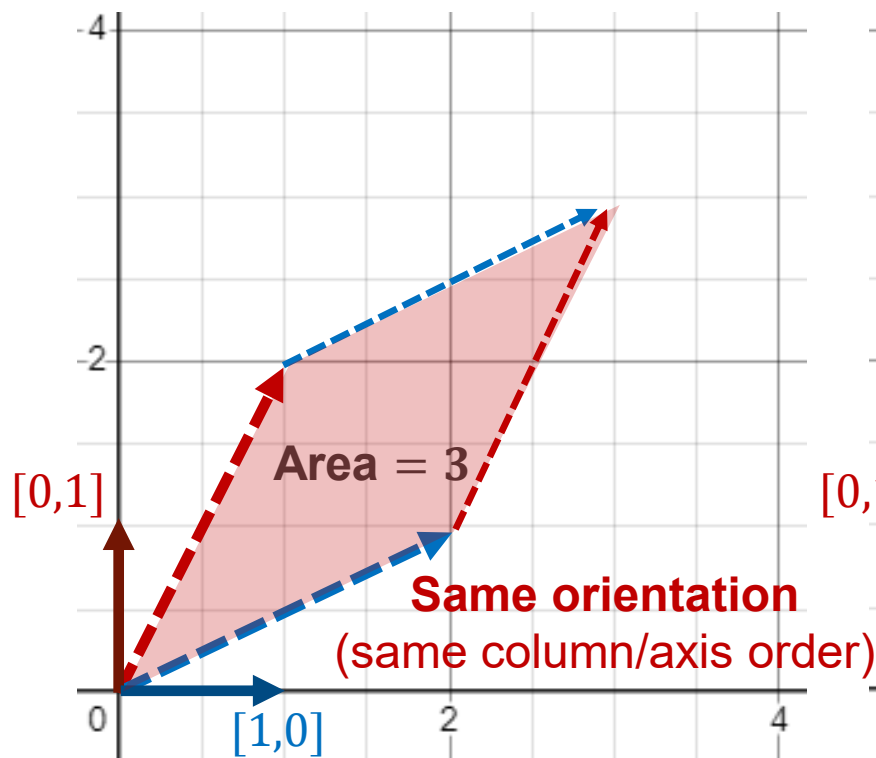
For shears

Matrix Determinant - Meaning (III)

$$\det A = |A| = a_{1,1}a_{2,2} - a_{1,2}a_{2,1}$$

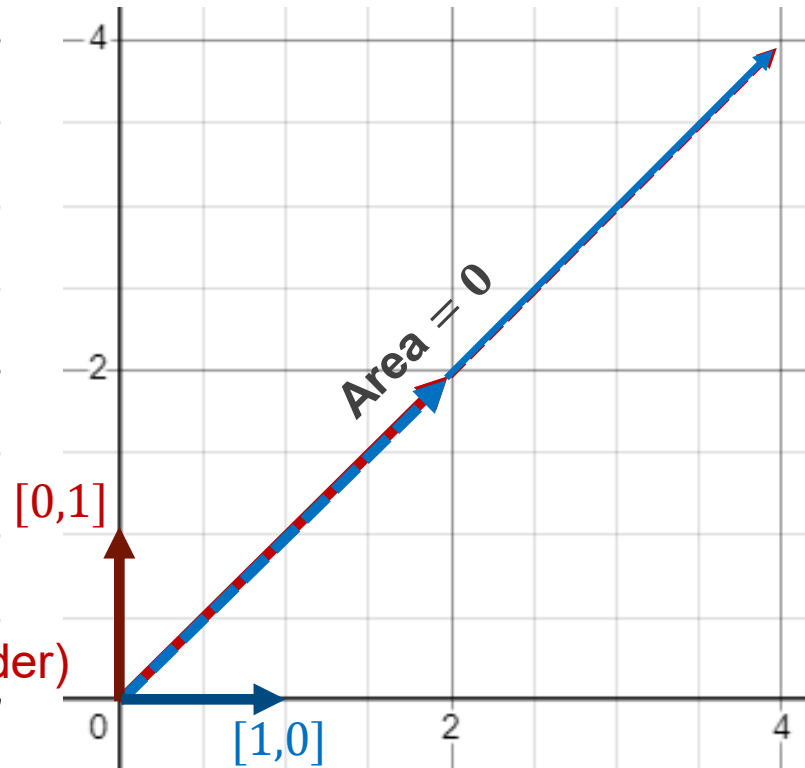
$$A = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}$$

$$\det A = 4 - 1 = 3$$



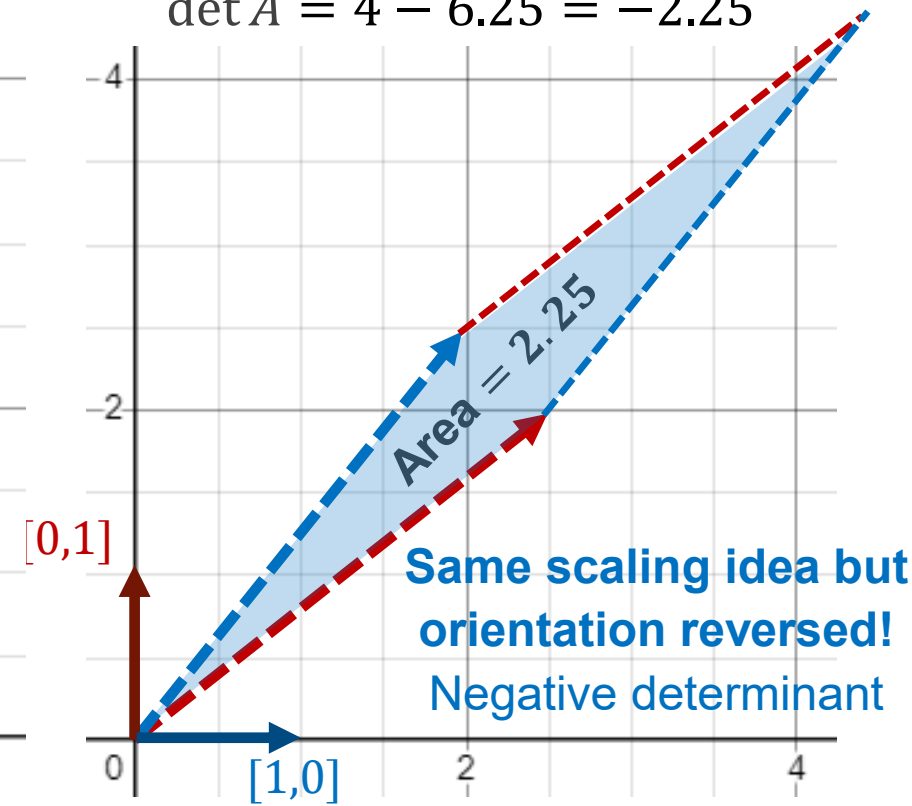
$$A = \begin{bmatrix} 2 & 2 \\ 2 & 2 \end{bmatrix}$$

$$\det A = 4 - 4 = 0$$



$$A = \begin{bmatrix} 2 & 2.5 \\ 2.5 & 2 \end{bmatrix}$$

$$\det A = 4 - 6.25 = -2.25$$



Note: the **span** of the vectors is the same (they are even the same)

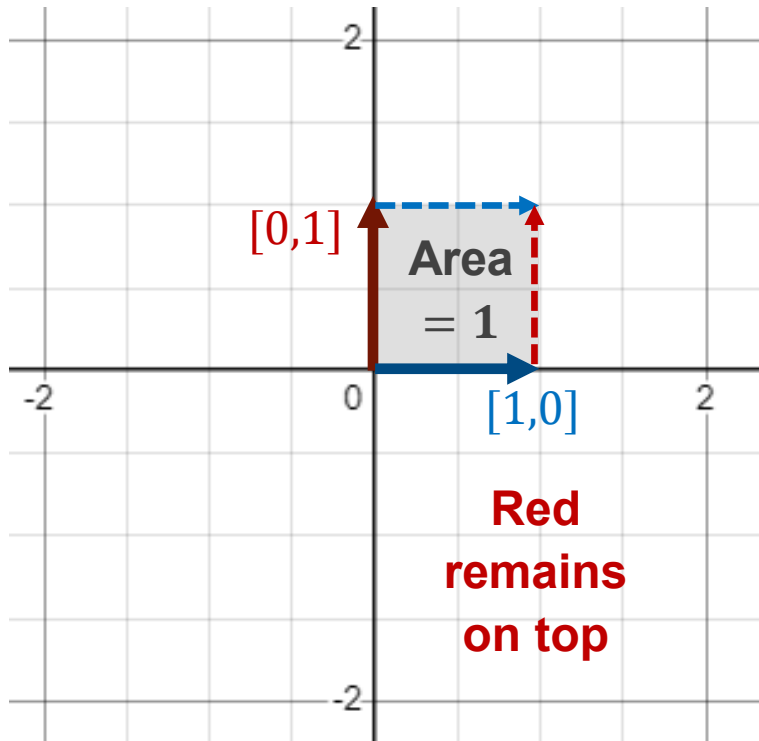


Change of orientation

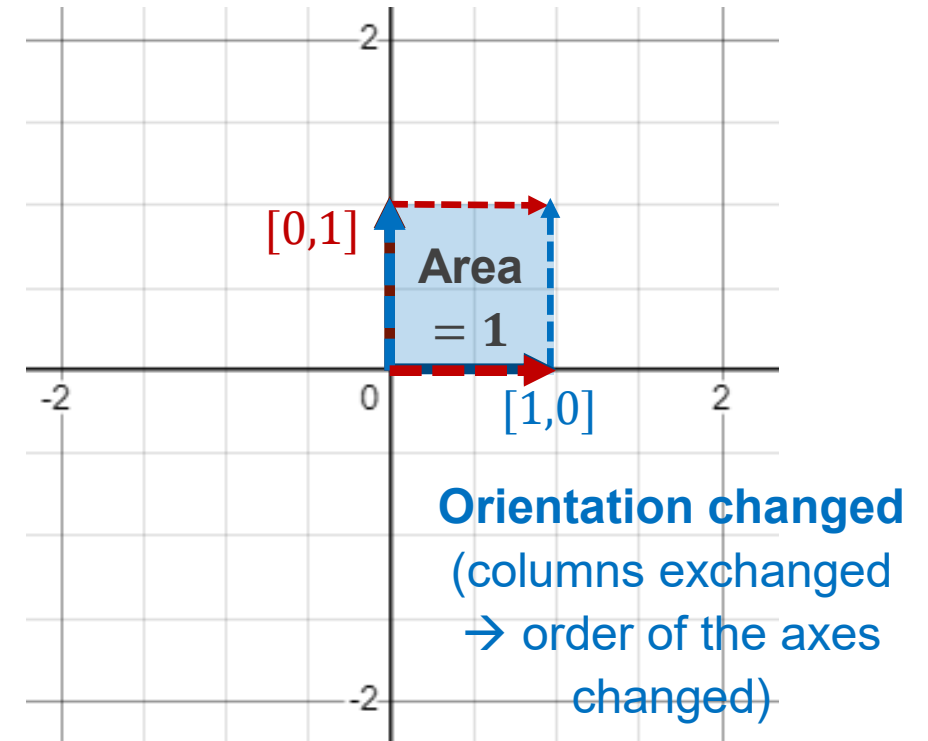
Matrix Determinant - Meaning (IV)

$$\det A = |A| = a_{1,1}a_{2,2} - a_{1,2}a_{2,1}$$

$$A = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \quad \det A = 0 - 1 = -1$$



← Standard generators
no transformation yet

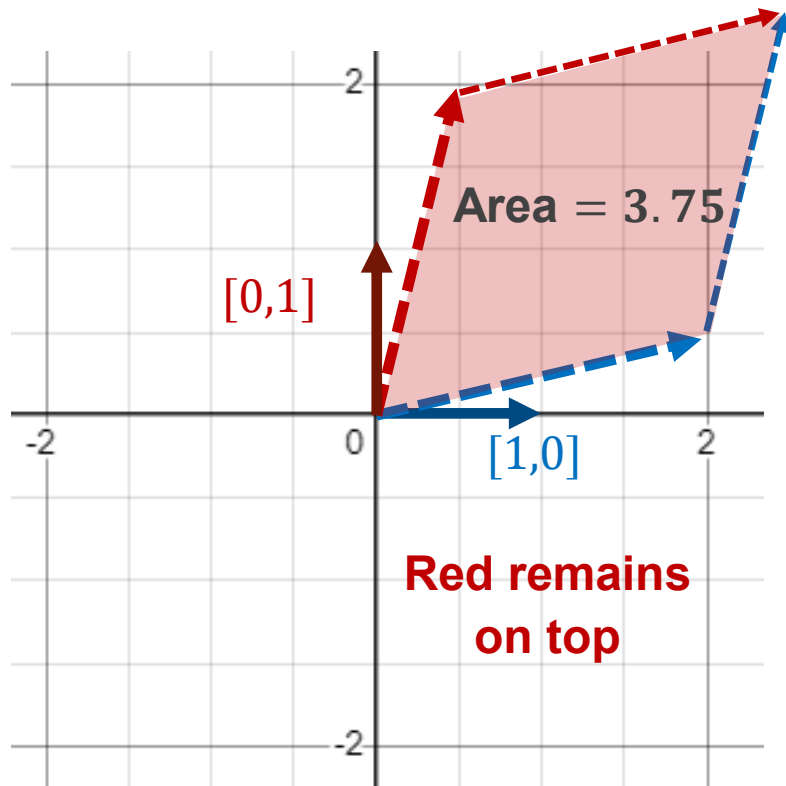




Change of orientation

Matrix Determinant - Meaning (II)

$$A = \begin{bmatrix} 2 & 0.5 \\ 0.5 & 2 \end{bmatrix} \quad \det A = 3.75$$

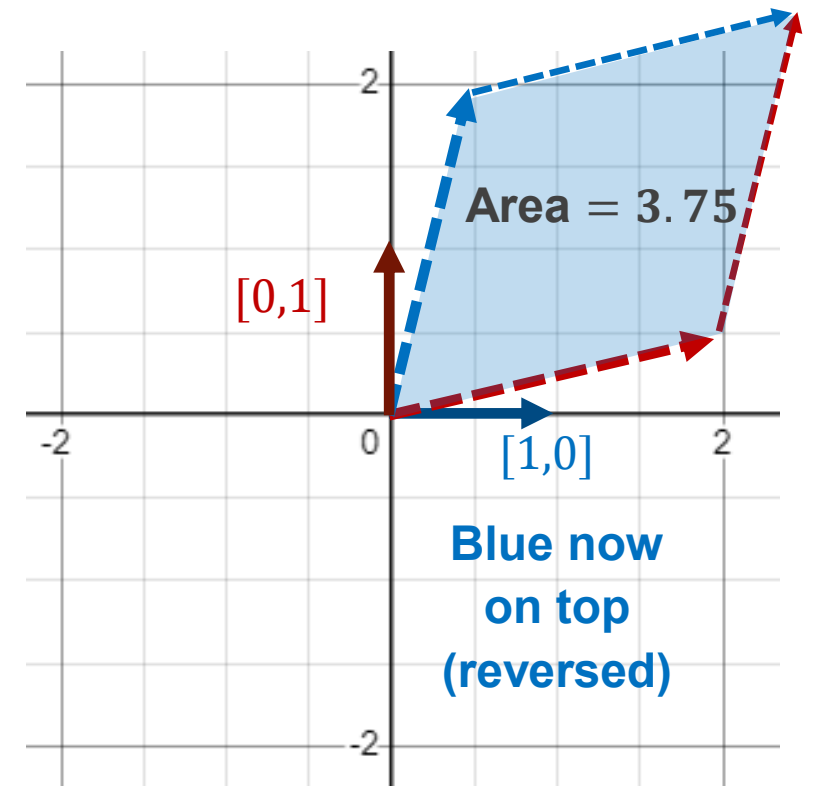


Orientation changes
(as indicated by the sign)



$$A = \begin{bmatrix} 0.5 & 2 \\ 2 & 0.5 \end{bmatrix} \quad \det A = -3.75$$

(since flipped)



For scaling - negative

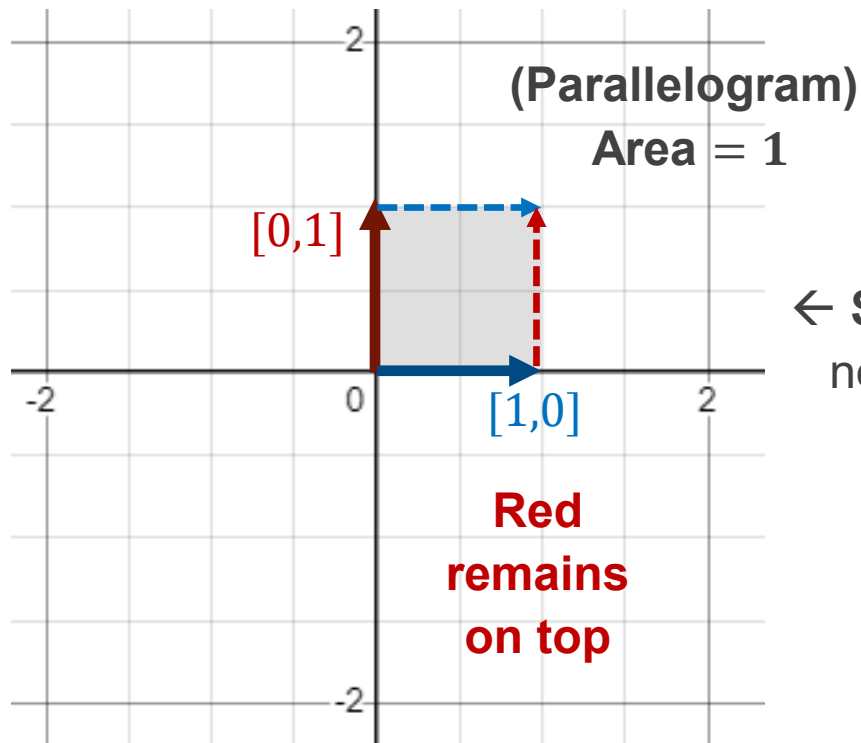
Matrix Determinant - Meaning (V)

Second part is currently 0

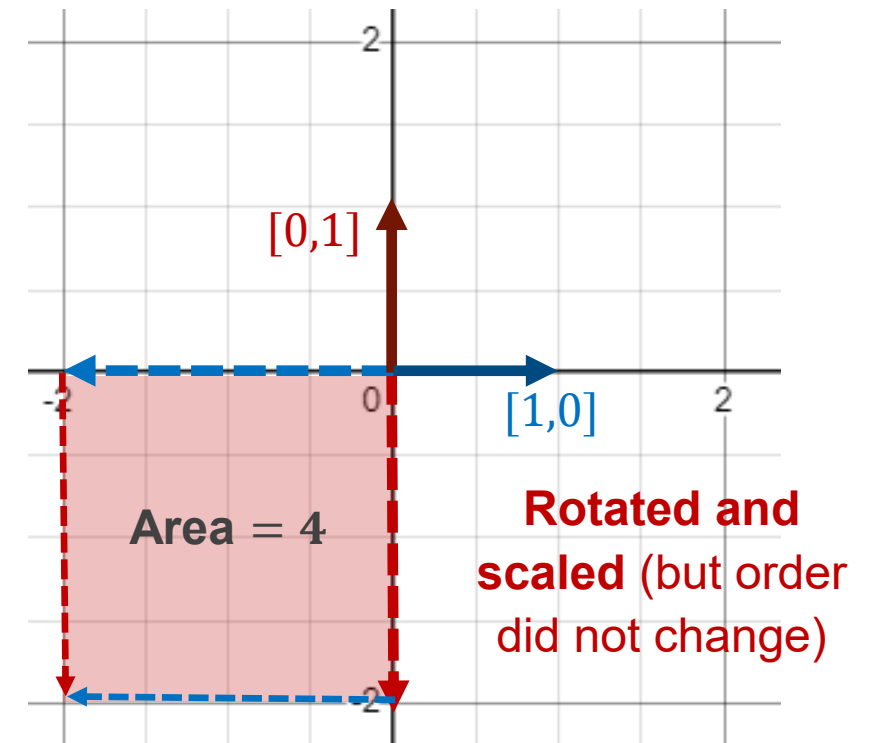


$$\det A = |A| = a_{1,1}a_{2,2} - a_{1,2}a_{2,1}$$

$$A = \begin{bmatrix} -2 & 0 \\ 0 & -2 \end{bmatrix} \quad \det A = 4 - 0 = 4$$



← Standard generators
no transformation yet





For shears- negative

Matrix Determinant - Meaning (VI)

$$A = \begin{bmatrix} -2 & 0.5 \\ 0.5 & -2 \end{bmatrix}$$

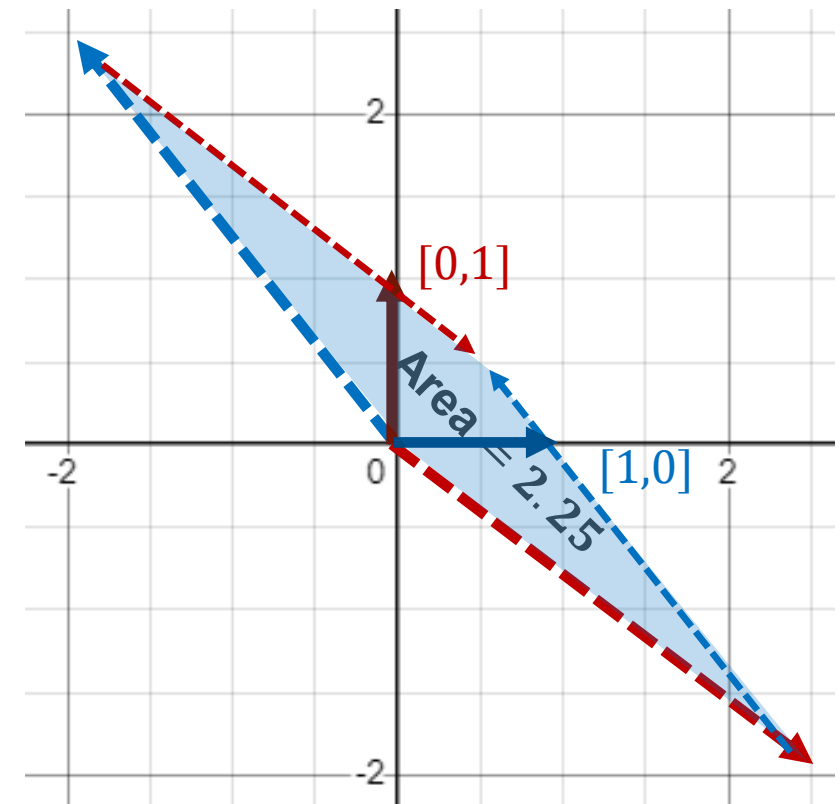
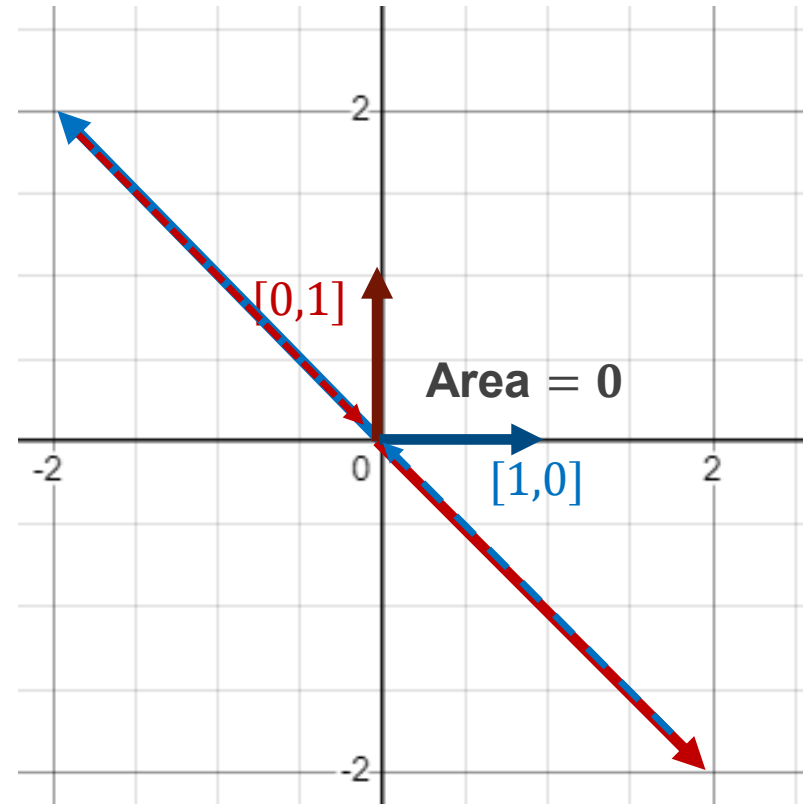
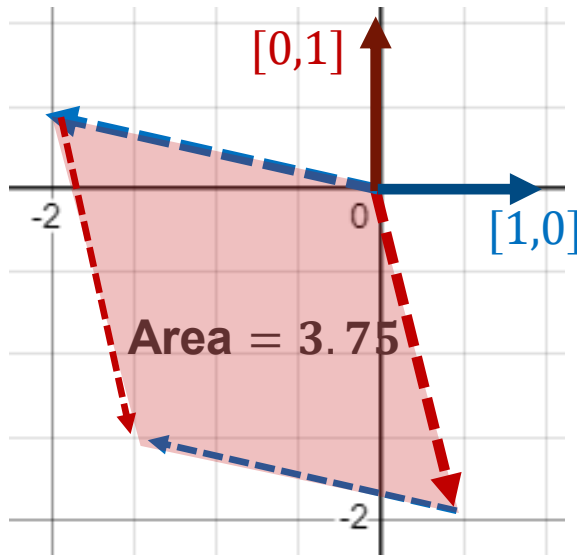
$$\det A = 4 - 0.25 = 3.75$$

$$A = \begin{bmatrix} -2 & 2 \\ 2 & -2 \end{bmatrix}$$

$$\det A = 4 - 4 = 0$$

$$A = \begin{bmatrix} -2 & 2.5 \\ 2.5 & -2 \end{bmatrix}$$

$$\det A = 4 - 6.25 = -2.25$$

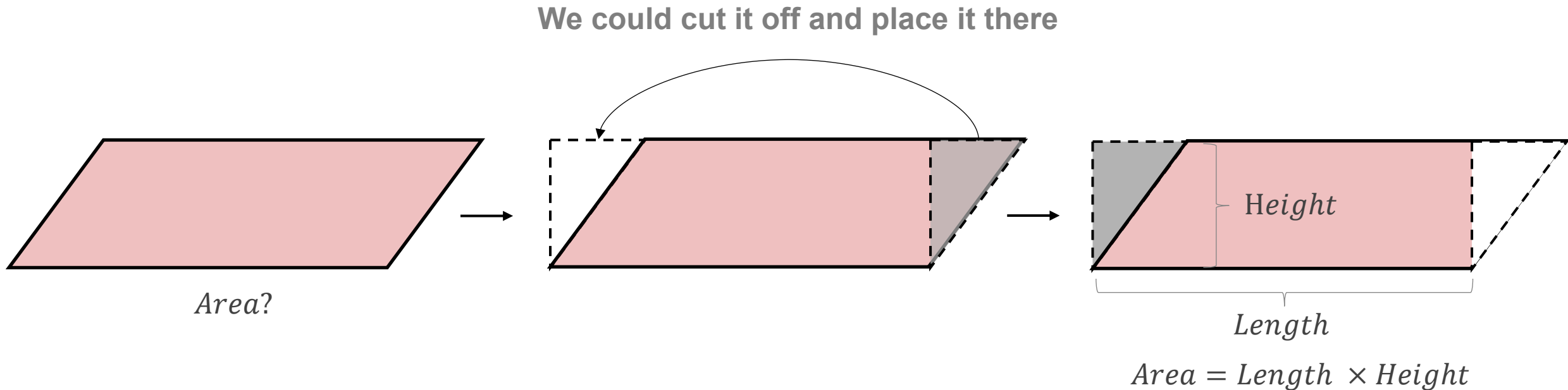




Matrix Determinant - Meaning (II)

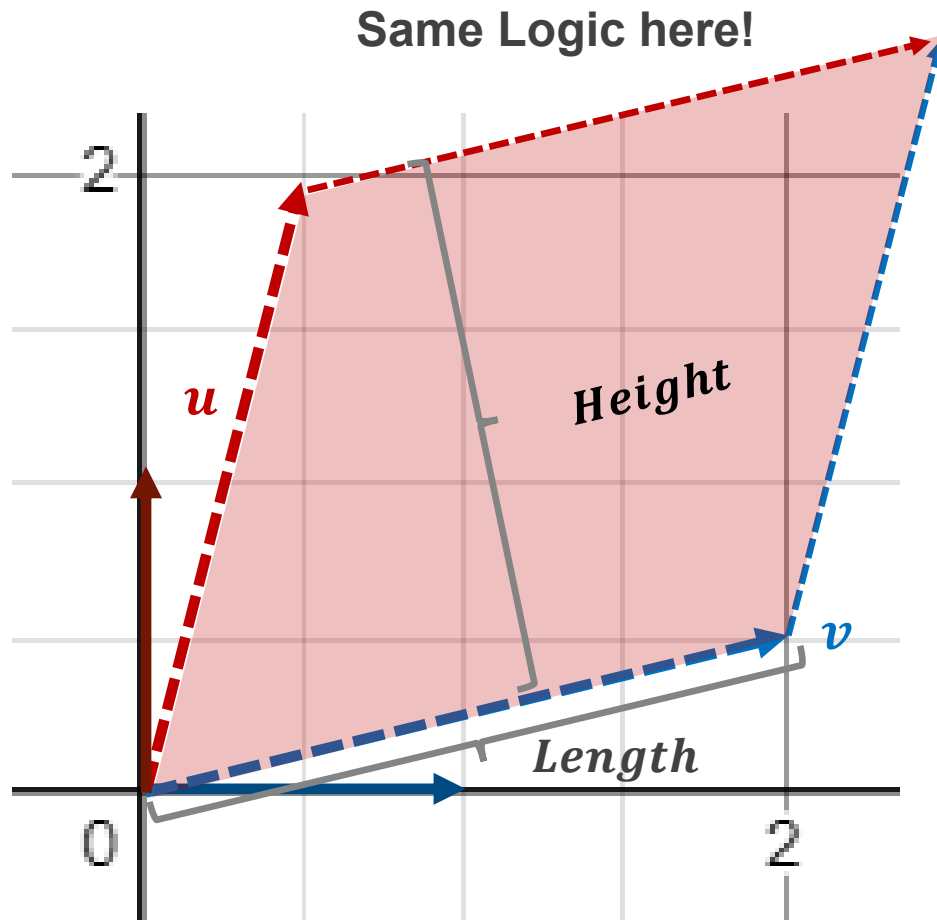
Why is that valuable to know for us?

One reason is to know about **independence** of the column vectors in a matrix





Matrix Determinant - Meaning (III)



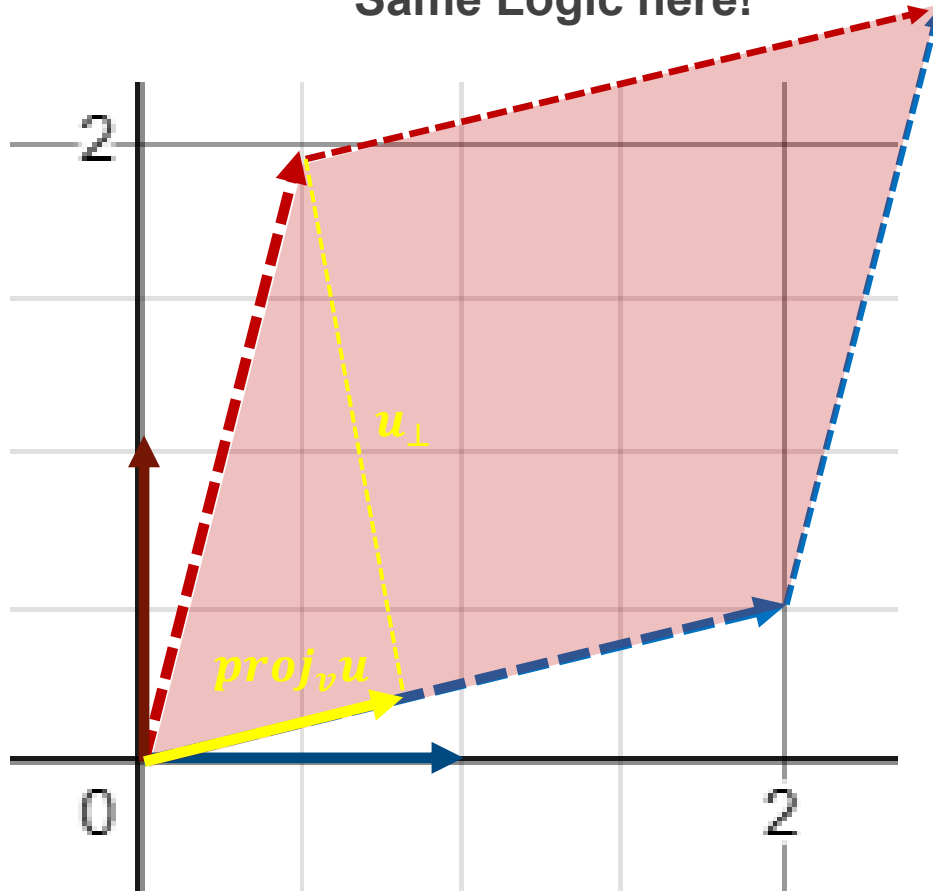
$$A = \begin{bmatrix} \textcolor{red}{u} & \textcolor{blue}{v} \\ 2 & 1 \\ 1 & 2 \end{bmatrix} \quad \det A = \textcolor{red}{3}$$

The length is simply the **length** of the vector. But what is the **height**?



Matrix Determinant - Meaning (III)

Same Logic here!



$$A = \begin{bmatrix} \textcolor{red}{2} & \textcolor{blue}{1} \\ \textcolor{red}{1} & \textcolor{blue}{2} \end{bmatrix} \quad \det A = \textcolor{red}{3}$$

As we will in later lectures see and be able to calculate, it is based on the projection of $\textcolor{red}{u}$ on $\textcolor{blue}{v}$.

Here:

$$\textcolor{blue}{Length} = \| \textcolor{blue}{v} \| = \sqrt{(1)^2 + (2)^2} = \sqrt{5}$$

$$\textcolor{yellow}{Height} = \| \textcolor{yellow}{u}_{\perp} \| = \sqrt{\left(-\frac{3}{5}\right)^2 + \left(\frac{6}{5}\right)^2} = \frac{3}{\sqrt{5}}$$

$$\textcolor{blue}{Length} \times \textcolor{yellow}{Height} = \sqrt{5} \times \frac{3}{\sqrt{5}} = 3$$





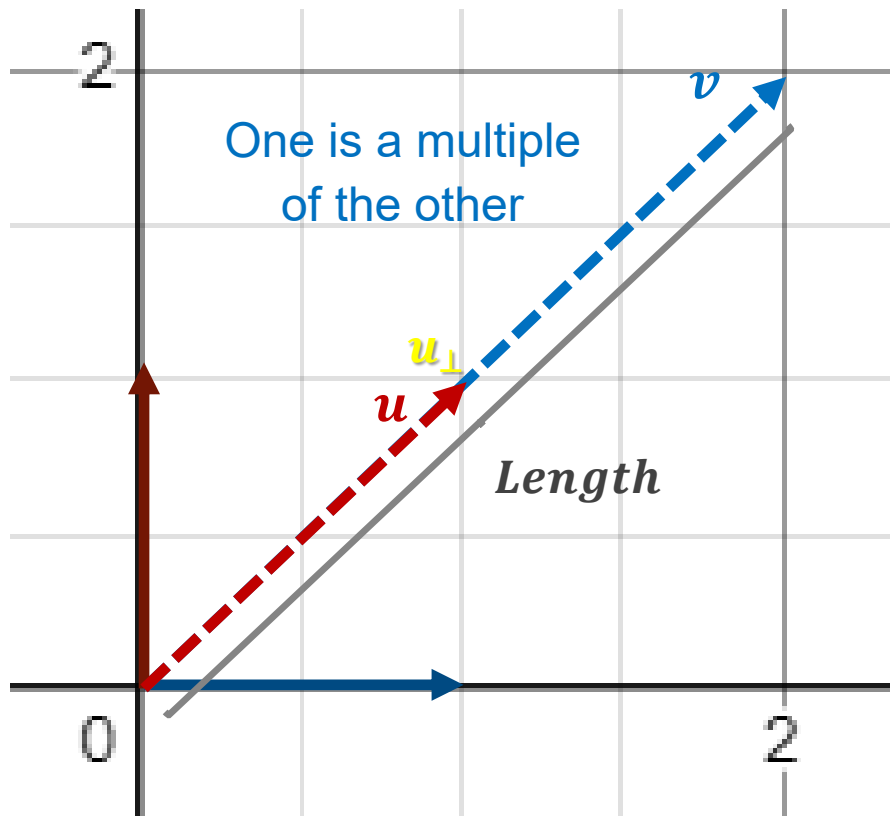
Matrix Determinant - Meaning (IV)

Different Example

u v

$$A = \begin{bmatrix} 1 & 2 \\ 1 & 2 \end{bmatrix} \quad \det A = 0$$

Now with columns
that have the same span



Think about dependence in terms of the span:
Does the one vector increase the span when adding it to the set containing the other vector?

Here:

$$\text{Length} = \|v\| = \sqrt{(2)^2 + (2)^2} = \sqrt{8}$$

$$\text{Height} = \|u_{\perp}\| = \sqrt{(0)^2 + (0)^2} = 0$$



$$\text{Length} \times \text{Height} = \sqrt{8} \times 0 = 0$$



Matrix Determinant - Calculation for 3x3

$$A = \begin{bmatrix} a_{1,1} & a_{1,2} \\ a_{2,1} & a_{2,2} \end{bmatrix} \quad \longrightarrow \quad \det A = |A| = a_{1,1}a_{2,2} - a_{1,2}a_{2,1}$$

$$A = \begin{bmatrix} a_{1,1} & a_{1,2} & a_{1,3} \\ a_{2,1} & a_{2,2} & a_{2,3} \\ a_{3,1} & a_{3,2} & a_{3,3} \end{bmatrix} \quad \longrightarrow \quad \det A = |A| = a_{1,1} \begin{vmatrix} a_{2,2} & a_{2,3} \\ a_{3,2} & a_{3,3} \end{vmatrix} - a_{1,2} \begin{vmatrix} a_{2,1} & a_{2,3} \\ a_{3,1} & a_{3,3} \end{vmatrix} + a_{1,3} \begin{vmatrix} a_{2,1} & a_{2,2} \\ a_{3,1} & a_{3,2} \end{vmatrix}$$

$$= a_{1,1}(a_{2,2}a_{3,3} - a_{2,3}a_{3,2}) - a_{1,2}(a_{2,1}a_{3,3} - a_{2,3}a_{3,1}) + a_{1,3}(a_{2,1}a_{3,2} - a_{2,2}a_{3,1})$$

Think of it as the signed ($\rightarrow$ orientation) **volume of the parallelepiped** (i.e., 3-dimensional analogue of a parallelogram)

$$\begin{bmatrix} a_{1,1} & a_{1,2} & a_{1,3} \\ a_{2,1} & a_{2,2} & a_{2,3} \\ a_{3,1} & a_{3,2} & a_{3,3} \end{bmatrix}$$

Submatrix

$$\begin{bmatrix} a_{1,1} & a_{1,2} & a_{1,3} \\ a_{2,1} & a_{2,2} & a_{2,3} \\ a_{3,1} & a_{3,2} & a_{3,3} \end{bmatrix}$$

Submatrix

$$\begin{bmatrix} a_{1,1} & a_{1,2} & a_{1,3} \\ a_{2,1} & a_{2,2} & a_{2,3} \\ a_{3,1} & a_{3,2} & a_{3,3} \end{bmatrix}$$

Submatrix



Matrix Determinant - Example & Dependence (I)

$$A = \begin{bmatrix} a_{1,1} & a_{1,2} & a_{1,3} \\ a_{2,1} & a_{2,2} & a_{2,3} \\ a_{3,1} & a_{3,2} & a_{3,3} \end{bmatrix} \Rightarrow \det A = |A| = a_{1,1} \begin{vmatrix} a_{2,2} & a_{2,3} \\ a_{3,2} & a_{3,3} \end{vmatrix} - a_{1,2} \begin{vmatrix} a_{2,1} & a_{2,3} \\ a_{3,1} & a_{3,3} \end{vmatrix} + a_{1,3} \begin{vmatrix} a_{2,1} & a_{2,2} \\ a_{3,1} & a_{3,2} \end{vmatrix}$$

$$\begin{bmatrix} a_{1,1} & a_{1,2} & a_{1,3} \\ a_{2,1} & a_{2,2} & a_{2,3} \\ a_{3,1} & a_{3,2} & a_{3,3} \end{bmatrix}$$

Submatrix

$$\begin{bmatrix} a_{1,1} & a_{1,2} & a_{1,3} \\ a_{2,1} & a_{2,2} & a_{2,3} \\ a_{3,1} & a_{3,2} & a_{3,3} \end{bmatrix}$$

Submatrix

$$\begin{bmatrix} a_{1,1} & a_{1,2} & a_{1,3} \\ a_{2,1} & a_{2,2} & a_{2,3} \\ a_{3,1} & a_{3,2} & a_{3,3} \end{bmatrix}$$

Submatrix

An example with
independent
columns

$$A = \begin{bmatrix} 1 & 1 & 1 \\ 2 & -2 & 2 \\ 3 & 3 & -3 \end{bmatrix}$$

$$\det A = 1 \begin{vmatrix} -2 & 2 \\ 3 & -3 \end{vmatrix} - 1 \begin{vmatrix} 2 & 2 \\ 3 & -3 \end{vmatrix} + 1 \begin{vmatrix} 2 & -2 \\ 3 & 3 \end{vmatrix}$$

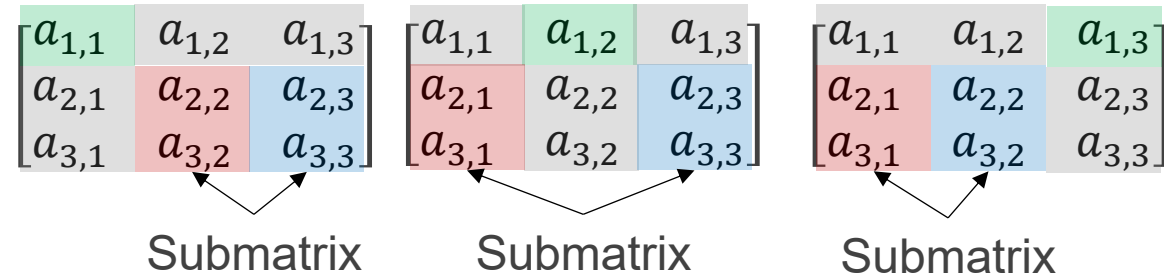
$$= 1(6 - 6) - 1(-6 - 6) + 1(6 - (-6))$$

$$= 0 - (-12) + 12$$

$$= 24$$



Matrix Determinant - Example & Dependence (II)



An example with
one dependent
column

$$A = \begin{bmatrix} 1 & 1 & 2 \\ 2 & -2 & 0 \\ 3 & 3 & 6 \end{bmatrix}$$



$$\begin{aligned} \det A &= 1 \begin{vmatrix} -2 & 0 \\ 3 & 6 \end{vmatrix} - 1 \begin{vmatrix} 2 & 0 \\ 3 & 6 \end{vmatrix} + 2 \begin{vmatrix} 2 & -2 \\ 3 & 3 \end{vmatrix} \\ &= 1(-12 - 0) - 1(12 - 0) + 2(6 - (-6)) = 0 \end{aligned}$$

An example with
all dependent
column

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 2 & 4 & 6 \\ 3 & 6 & 9 \end{bmatrix}$$



$$\begin{aligned} \det A &= 1 \begin{vmatrix} 4 & 6 \\ 6 & 9 \end{vmatrix} - 2 \begin{vmatrix} 2 & 6 \\ 3 & 9 \end{vmatrix} + 3 \begin{vmatrix} 2 & 4 \\ 3 & 6 \end{vmatrix} \\ &= 1(36 - 36) - 2(18 - 18) + 3(12 - 12) = 0 \end{aligned}$$



Matrix Determinant - Example & Dependence (III)

**Independent
columns**

$$A = \begin{bmatrix} 1 & 1 & 1 \\ 2 & -2 & 2 \\ 3 & 3 & -3 \end{bmatrix}$$

$$\det A = 24$$

Non-zero

A matrix with a **non-zero** determinant has independent columns (and is **invertible**)

**One dependent
column**

$$A = \begin{bmatrix} 1 & 1 & 2 \\ 2 & -2 & 0 \\ 3 & 3 & 6 \end{bmatrix}$$

$$\det A = 0$$

Zero

A matrix with a **zero** determinant has one or more dependent columns (and is **not invertible**)

Intuition: Think of it as volume based on Height x (Length x Width) where one or more component(s) are zero – the product would be zero

**All dependent
column**

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 2 & 4 & 6 \\ 3 & 6 & 9 \end{bmatrix}$$

$$\det A = 0$$

Zero



Matrix Inverse (I)



Remember (Lecture 1)

Definition: Functions f and g are *functional inverses* if $f \circ g$ and $g \circ f$ are defined and are identity functions.

A function that has an inverse is *invertible*.

The two functions ...

$$f(x) = Ax$$

$$g(x) = Bx$$

... are functional inverses...

$$(g \circ f)(x) = x$$

$$(f \circ g)(x) = x$$



Matrix Inverse (II)

The two functions ...

$$f(x) = Ax$$

$$g(x) = Bx$$

... are functional inverses...

$$(g \circ f)(x) = x$$

$$(f \circ g)(x) = x$$

...if we plug in the
underlying matrices
and solve:

$$B(Ax) = x$$

$$A(Bx) = x$$

$$BAx = x$$

$$ABx = x$$

$$\rightarrow BA = I$$

$$\rightarrow AB = I$$

Definition If f and g are functional inverses of each other, we say A and B are matrix inverses of each other.



Matrix Inverse (III)

If A and B are matrix inverses of each other, we say A and B are invertible matrices.

We denote the inverse of matrix A by A^{-1} .

(A matrix that is not invertible is sometimes called a singular matrix, and an invertible matrix is called a *nonsingular* matrix.)



Matrix Inverse (IV)

Corollary: A and B are inverses of each other iff both AB and BA are identity matrices.

Remember (Lecture 1)

Definition: Functions f and g are *functional inverses* if $f \circ g$ and $g \circ f$ are defined and are identity functions.

Already in Lecture 1 when talking about functional inverses, it had to “work” both ways and result in the identity (function)



Matrix Inverse - Importance

Consider the matrix-vector equation $Ax = b$

If A has an inverse A^{-1} then by multiplying both sides of the equation on the left by A^{-1} , we obtain the equation

$$A^{-1}Ax = A^{-1}b$$

Where $A^{-1}A$ is the identity matrix I , which multiplied by x results in x .

$$x = A^{-1}b$$

This tells us that if equation $Ax = b$ has a solution, then the solution must be $x = A^{-1}b$.
(if A is invertible) (unique solution)



Product of Invertible Matrices (I)

If individual
matrices invertible

A and B



Then individual
functions invertible

$f(x) = Ax$ and $g(x) = Bx$.



Then the
composition of these
functions invertible

$f \circ g$



Then corresponding
matrix product
invertible

$(f \circ g)(x) = (AB)x$





Product of Invertible Matrices (II)

Lemma: If the $R \times C$ matrix A has an inverse A^{-1} then AA^{-1} is identity matrix.

What about the converse?

False

Conjecture: If AB is an identity matrix then A and B are inverses...?

But if I did something and then take another action and get back to the beginning ($\rightarrow$ identity matrix), could I not have messed it up some other way ($\rightarrow$ not necessarily inverse)



But also





Product of Invertible Matrices (III)

Counterexample:

$$A = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix}, B = \begin{bmatrix} 1 & 0 \\ 0 & 1 \\ 0 & 0 \end{bmatrix}$$

$$AB = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 0 & 1 \\ 0 & 0 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

$A * [0, 0, 1]$ and $A * [0, 0, 0]$ both equal $[0, 0]$, so null space of A is not trivial, so the function $f(\mathbf{x}) = A\mathbf{x}$ is not one-to-one, so f is not an invertible function.

$$A * \begin{bmatrix} 1 & 0 \\ 0 & 1 \\ 5 & 4 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

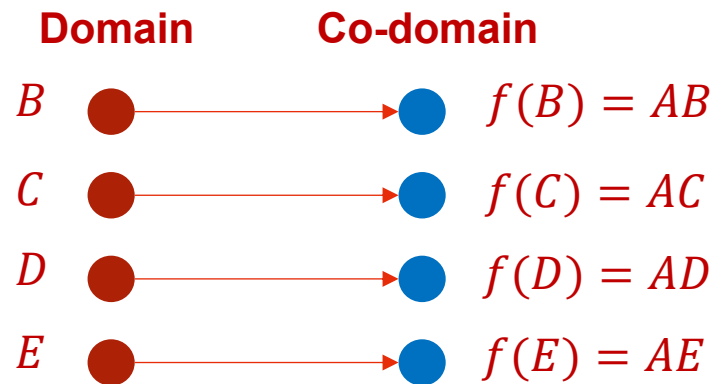
Shows: $AB = I$ is *not* sufficient to ensure that A and B are inverses.



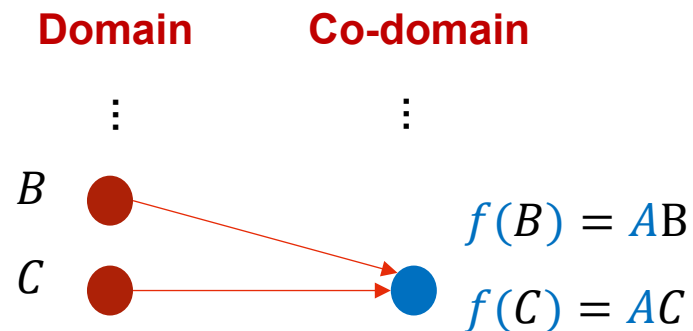


Product of Invertible Matrices (IV)

How one-to-one would look



But unfortunately



$$A = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix} \quad B = \begin{bmatrix} 1 & 0 \\ 0 & 1 \\ 0 & 0 \end{bmatrix} \quad C = \begin{bmatrix} 1 & 0 \\ 0 & 1 \\ 5 & 4 \end{bmatrix}$$

$$AB = A * \begin{bmatrix} 1 & 0 \\ 0 & 1 \\ 0 & 0 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

$$AC = A * \begin{bmatrix} 1 & 0 \\ 0 & 1 \\ 5 & 4 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$



(and there are more inputs for this example that would result in the identity matrix)



Exercise: Determinants & Inverses

You are given the following matrices:

$$A = \begin{bmatrix} 1 & 2 \\ 1 & 1 \end{bmatrix} \quad B = \begin{bmatrix} -1 & 2 \\ 1 & -1 \end{bmatrix} \quad C = \begin{bmatrix} 2 & 1 & 1 \\ -1 & 2 & -3 \\ 2 & 3 & -1 \end{bmatrix}$$

- 1) What is the determinant of Matrix A ? What does that tell you about the columns being independent and the matrix being invertible? Explain.
- 2) What is the determinant of Matrix B . Show work.
- 3) Based on the results above, will the matrix product of $A \times B$ be invertible? Explain.
- 4) Calculate $A \times B$. Based on just that result, what can you conclude about A and B being inverses?
- 5) What is the determinant of Matrix C ?

Solution**Exercise: Determinant**

$$A = \begin{bmatrix} 1 & 2 \\ 1 & 1 \end{bmatrix} \quad B = \begin{bmatrix} -1 & 2 \\ 1 & -1 \end{bmatrix} \quad C = \begin{bmatrix} 2 & 1 & 1 \\ -1 & 2 & -3 \\ 2 & 3 & -1 \end{bmatrix}$$

a) $\det A = 1 \times 1 - 2 \times 1 = -1$

Since the determinant is nonzero and the matrix is invertible (and the columns of A are independent)

b) $\det B = (-1) \times (-1) - 2 \times 1 = -1$

c) Since for both matrices, the determinant is nonzero, the matrices are invertible. The product of two invertible matrices is also invertible.

d) $A \times B = \begin{bmatrix} 1 & 2 \\ 1 & 1 \end{bmatrix} \times \begin{bmatrix} -1 & 2 \\ 1 & -1 \end{bmatrix} = \begin{bmatrix} 1 \times (-1) + 2 \times 1 & 1 \times 2 + 2 \times (-1) \\ 1 \times (-1) + 1 \times 1 & 1 \times 2 + 1 \times (-1) \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$

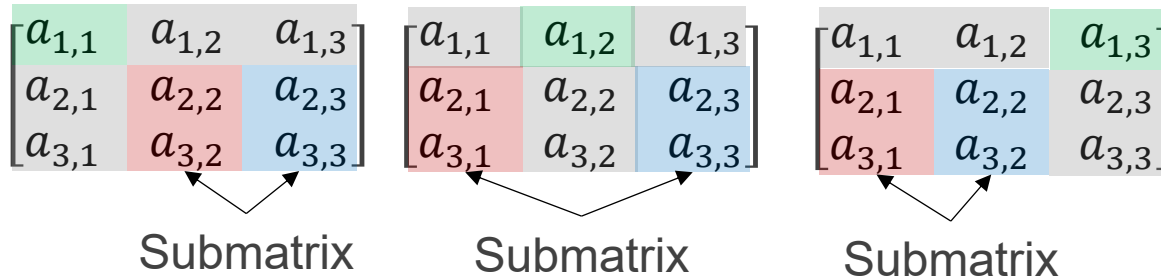
With our current knowledge (this far into the course), we cannot conclude that A and B are inverses by only knowing $AB = I$. We would also need to know that $BA = I$ which we have not checked yet.

Solution

Exercise: Determinant

$$A = \begin{bmatrix} 1 & 2 \\ 1 & 1 \end{bmatrix} \quad B = \begin{bmatrix} -1 & 2 \\ 1 & -1 \end{bmatrix} \quad C = \begin{bmatrix} 2 & 1 & 1 \\ -1 & 2 & -3 \\ 2 & 3 & -1 \end{bmatrix}$$

e)



$$\begin{aligned} \det C &= 2 \begin{vmatrix} 2 & -3 \\ 3 & -1 \end{vmatrix} - 1 \begin{vmatrix} -1 & -3 \\ 2 & -1 \end{vmatrix} + 1 \begin{vmatrix} -1 & 2 \\ 2 & 3 \end{vmatrix} \\ &= 2(-2 - (-9)) - 1(1 - (-6)) + 1(-3 - 4) \\ &= 2(7) - 1(7) + 1(-7) = 0 \end{aligned}$$

Alternatively: If one is able to spot that column 3 is simply column 1 – column 2, one can conclude that the columns are not independent. Thus, the determinant would reflect that with a value of 0 – showing that the columns are dependent and that the matrix is not invertible.



Some Key Take-aways (Lecture 10)

A vector can be interpreted as a **column vector** meaning a one-column matrix or a **row vector** meaning a one-row matrix

Previously only
considered an n-vector

The **matrix-multiplication lemma** states that the composition of two functions $f(y) = A * y$ and $g(x) = B * x$ is the function $h(x) = (AB) * x$

The inverse of a Matrix A is A^{-1} and if A is **invertible**, then $f(x) = A * x$ is an invertible function, meaning it is one-to-one and onto $\rightarrow$ for $A * x = b$ a **solution exists** as it is onto and there is **at most one vector** to b as it is one-to-one

A solution exists
and it is unique

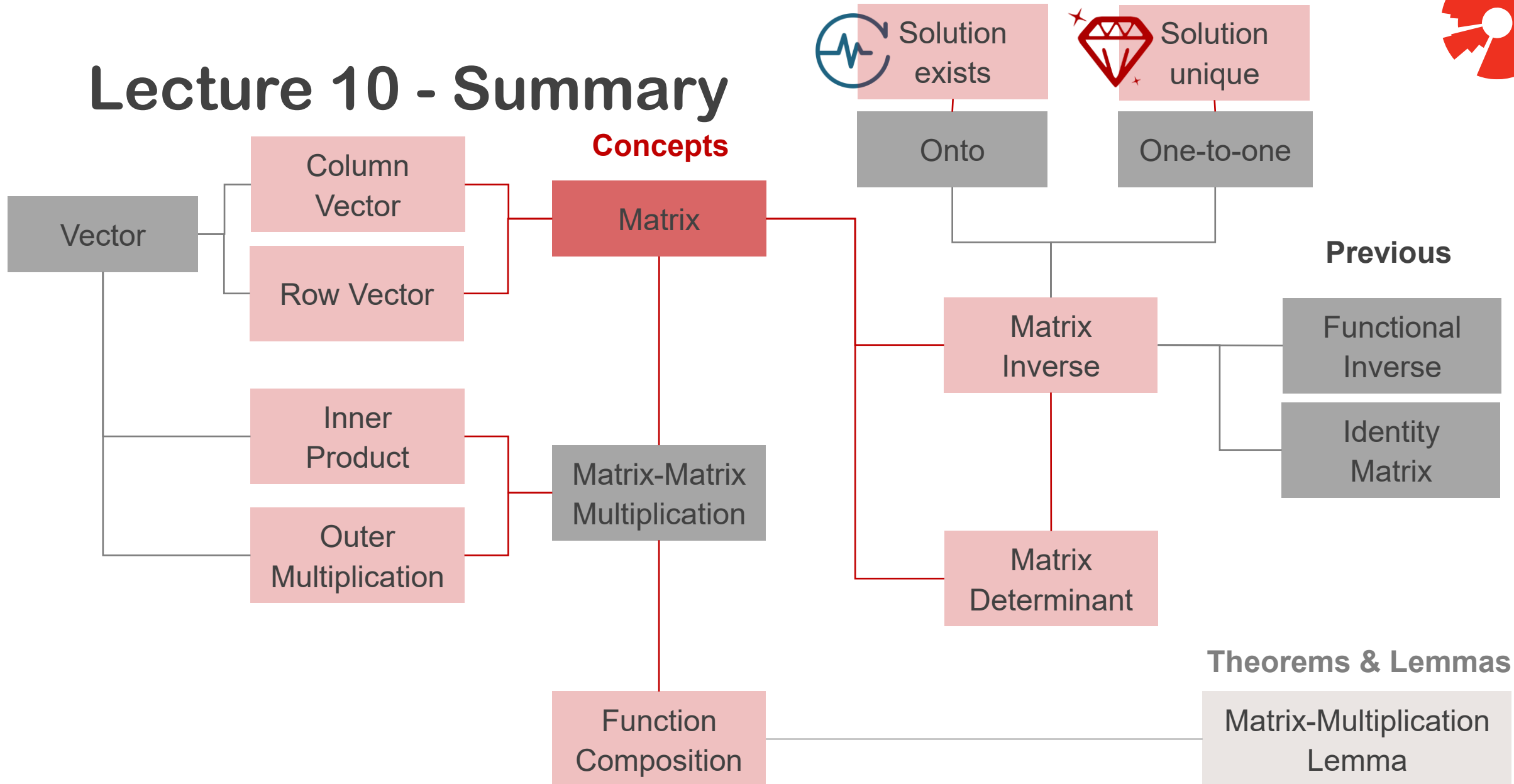
The **matrix determinant** provides one way of knowing if a matrix is invertible

A vector $v = \alpha_1 a_1 + \dots + \alpha_n a_n$ can be presented via the vector of coefficients $[\alpha_1, \dots, \alpha_n]$ called the **coordinate representation** of v in terms of $a_1, \dots, a_n$, which can be represented via a **matrix-vector equation** $Au = v$

One can also do
this also for
standard generators



Lecture 10 - Summary





Thank You For Your Attention!



**Accomplishment unlocked:
Lecture 10 Completed**



APPENDIX



Matrix Inverse - Importance (II)

Reason 1: Existence and uniqueness of solution to matrix-vector equations.

Let A be an $m \times n$ matrix, and define $f : \mathbb{F}^n \longrightarrow \mathbb{F}^m$ by $f(\mathbf{x}) = A\mathbf{x}$

Suppose A is an invertible matrix. Then f is an invertible function. Then f is one-to-one and onto:



- ▶ Since f is onto, for any m -vector $\mathbf{b}$ there is *some* vector $\mathbf{u}$ such that $f(\mathbf{u}) = \mathbf{b}$. That is, there is at least one solution to the matrix-vector equation $A\mathbf{x} = \mathbf{b}$.



- ▶ Since f is one-to-one, for any m -vector $\mathbf{b}$ there is *at most one* vector $\mathbf{u}$ such that $f(\mathbf{u}) = \mathbf{b}$. That is, there is at most one solution to $A\mathbf{x} = \mathbf{b}$.

If A is invertible then, for every right-hand side vector $\mathbf{b}$, the equation $A\mathbf{x} = \mathbf{b}$ has *exactly one solution*.



Matrix Inverse - Importance (III)



EASY

Reason 2: Algorithms for solving matrix-vector equation $A\mathbf{x} = \mathbf{b}$ are simpler if we can assume A is invertible.

Later we learn two such algorithms.

We also learn how to cope if A is not invertible.

Reason 3:

Invertible matrices play a key role in *change of basis*.



Change of basis is important part of linear algebra

- ▶ used e.g. in image compression;
- ▶ we will see it used in adding/removing perspective from an image.



Product of Invertible Matrices (V)

Product of invertible matrices is also an invertible matrix

Proposition: Suppose the matrix product AB is defined. Then AB is an invertible matrix if and only if both A and B are invertible matrices.

Example:

$$f : \mathbb{R}^2 \longrightarrow \mathbb{R}^2 \text{ and } g : \mathbb{R}^2 \longrightarrow \mathbb{R}^2$$

$$A = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix} \text{ and } B = \begin{bmatrix} 1 & 0 \\ 1 & 1 \end{bmatrix} \text{ correspond to functions}$$

Example continues next slide ...



Product of Invertible Matrices (VI)

$$f\left(\begin{bmatrix} x_1 \\ x_2 \end{bmatrix}\right) = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} x_1 + x_2 \\ x_2 \end{bmatrix}$$

$$g\left(\begin{bmatrix} x_1 \\ x_2 \end{bmatrix}\right) = \begin{bmatrix} 1 & 0 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} x_1 \\ x_1 + x_2 \end{bmatrix}$$

f is an invertible function.

g is an invertible function.

The functions f and g are invertible so the function $f \circ g$ is invertible.

By the Matrix-Multiplication Lemma, the function $f \circ g$ corresponds to the matrix

product $AB = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 1 & 1 \end{bmatrix} = \begin{bmatrix} 2 & 1 \\ 1 & 1 \end{bmatrix}$ so that matrix is invertible.



Product of Invertible Matrices (VII)

Example:

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix}$$

$$B = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 1 & 0 \end{bmatrix}$$

The product is $AB = \begin{bmatrix} 4 & 5 & 1 \\ 10 & 11 & 4 \\ 16 & 17 & 7 \end{bmatrix}$

which is not invertible

so at least one of A and B is not invertible

and in fact $\begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix}$ is not invertible.

So if one or multiple matrices in a matrix multiplication are not invertible, then the resulting matrix is also not invertible





Product of Invertible Matrices (VIII)

Proposition: Suppose the matrix product AB is defined. Then AB is an invertible matrix if and only both A and B are invertible matrices.



Proof: Define the functions f and g by $f(\mathbf{x}) = A\mathbf{x}$ and $g(\mathbf{x}) = B\mathbf{x}$.

- ▶ Suppose A and B are invertible matrices.
Then the corresponding functions f and g are invertible.
Therefore $f \circ g$ is invertible
so the matrix corresponding to $f \circ g$ (which is AB) is an invertible matrix.
- ▶ Conversely, suppose AB is an invertible matrix.
Then the corresponding function $f \circ g$ is an invertible function.
It follows that f and g must be invertible functions,
so the corresponding matrices A and B must be invertible matrices.

QED



Product of Invertible Matrices (IX)

Question: How can we tell if a matrix M is invertible?

Partial Answer: By definition, M is an invertible matrix if the function $f(\mathbf{x}) = M\mathbf{x}$ is an invertible function, i.e. if the function is one-to-one and onto.

- ✓ ▶ **One-to-one:** Since the function is linear, we know by the One-to-One Lemma that the function is one-to-one if its kernel is trivial, i.e. if the null space of M is trivial.
- ❓ ▶ **Onto:** We haven't yet answered the question *how we can tell if a linear function is onto?*

If we knew how to tell if a linear function is **onto**, therefore, we would know how to tell if a matrix is invertible.